

KALE PARNHAM

Mechanical Engineering student | University of New Brunswick

CONTACT

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Fredericton NB

MY PROFILES

LinkedIn: linkedin.com/in/tkalep
Portfolio: tkale.github.io/Portfolio/
GitHub: github.com/T-Kale

EDUCATION

BScE, Mechanical Engineering
University of New Brunswick
Expected graduation: 2029.

BScF, Forestry
University of New Brunswick
2022-2024

Charlottetown Rural High School

SKILLS

Design & Engineering:

- CAD - Fusion 360
- Structural Analysis (*Preliminary*)
- FEA (*Preliminary*)
- Additive Manufacturing (FDM)

Programming & Data Analysis:

- Excel
- Python

Field & Research:

- Data Collection
- Protocol Development

PROFESSIONAL SUMMARY

Mechanical Engineering student with experience leading structural design for competition robotics systems. Skilled in parametric CAD modeling, preliminary structural analysis, and subsystem integration using Fusion. Combines extensive hands-on field experience with disciplined problem-solving to deliver practical engineering solutions.

EXPERIENCE

Forest Technician – Dept. of Forests, Fish and Wildlife, Gov. of PEI
May 2024 – September 2024, May 2023 – September 2023

- Conducted large-scale field surveys across 1000+ acres to produce data for government research initiatives.
- Coordinated with stakeholders and contractors to develop and implement environmental management plans and best practices.

Invasive Species Tech – Prince Edward Island Watershed Alliance
July 2022 – August 2022

- Developed field protocols to identify and manage 13 invasive species.
- Collaborated with teams to evaluate experimental management strategies.
- Engaged in outreach efforts to improve awareness of ecological risks.

Watershed Field Tech – Kensington North Watershed Association
July 2021 – August 2021

- Stream restoration, invasive species removal, and wildlife surveys.
- Contributed field measurements to agricultural climate impact studies.

DESIGN PROJECTS

UNB Robotics Club / Canadian International Rover Challenge (CIRC)

- Led mechanical design for rover chassis and structural framework.
- Developed parametric assemblies and manufacturing drawings using Autodesk Fusion.
- Validated designs using preliminary hand calculations and FEA.
- Prototyped structural components using FDM 3D printing to verify fit, geometry, and assembly of components.
- Collaborated with internal mechanical and electrical teams to integrate critical hardware and additional subassemblies.

ACHIEVEMENTS

- Recipient: George and Elaine Memorial Scholarship in Forestry (2022)
- Recipient: UNB 75th Anniversary Scholarship in Forestry (2022)
- Honors recognition: Charlottetown Rural High School (2021)